

National Institute of Technology Patna
Department of Mathematics
Mid Semester Examination : May 2022

Course Name: **Engineering Mathematics-I**
Program: Civil Engineering(Batch-I).
Duration: 2 Hrs

Course Code: MA23101

Full Marks: 30

Answer all the questions

1. Reduce the matrix A to the normal form and hence find its rank where,

$$A = \begin{bmatrix} 2 & -4 & 3 & 1 & 0 \\ 1 & -2 & 1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \\ 4 & -7 & 4 & -4 & 5 \end{bmatrix}. \quad [4]$$

2. Examine the consistency of the following system of linear equations and solve it if consistent:

$$\begin{aligned} 2x_1 + x_2 - x_3 + 3x_4 &= 11 \\ x_1 - 2x_2 + x_3 + x_4 &= 8 \\ 4x_1 + 7x_2 + 2x_3 - x_4 &= 0 \\ 3x_1 + 5x_2 + 4x_3 + 4x_4 &= 17 \end{aligned}$$

[4]

3. Find the eigenvalues and corresponding eigenvectors the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$. [4]

4. Using Caley-Hamilton theorem for $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$, find A^{-1} and also find the matrix $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$. [4]

5. Is matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$, diagonalizable? If so, obtain the matrix P such that $P^{-1}AP$ is a diagonal matrix. [4]

6. Suppose the vectors u, v, w are linearly independent. Show that the vectors $u + v, u - v, u - 2v + w$ are also linearly independent. [3]

7. If W be the subspace of $\mathbb{R}^4$ generated by the vectors $(1, -2, 5, -3)$, $(2, 3, 1, -4)$, and $(3, 8, -3, -5)$, find a basis and the dimension of W . Extend the basis of W to a basis of $\mathbb{R}^4$. [4]

8. Solve the initial value problem

$$e^x(\cos y dx - \sin y dy) = 0, \quad y(0) = 0.$$

[3]

All the Best
